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Next >

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inverse times A times X is equal to and AX. Alright? Hmm, we can rewrite that as

A times X is equal to X times lambda. Now, what do we like to do in this course?

Well we like to take matrices and bust them up by columns. And

we already talked about the fact that X was the matrix where its columns were the basis vectors that we were using to express our vector in the new basis. So

let's think our X and let's partition it into its columns and let's

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Reading assignment

1/1 point (graded)

Read Unit 9.1.1 of the notes. [\[LINK\]](#)

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🗨 [Error when running showeig.m](#) 3

[After entering the A matrix and running the showeig.m, I got the following er...](#)

1/1 point (graded)
Eigenvalues and eigenvectors are all about finding scalars, λ , and nonzero vectors, x , such that

$$Ax = \lambda x.$$

To help you visualizing how a 2×2 real-valued matrix transforms a vector on the unit circle in general, and eigenvectors of unit length in particular, we have created the function

`Assignments/Week09/matlab/showeig.m` (inspired by such a function that used to be part of Matlab). You may want to do a "git pull" to update your local copy of the `Assignments` directory.

In the command window, first create a 2×2 matrix in array A and then execute `showeig( A )`.

Here are some matrices to try:

```
A = [ 2      0
      0 -0.5 ]
```

```
A = [ 2      1
      0 -0.5 ]
```

```
A = [ 2      1
      1 -0.5 ]
```

```
theta = pi/4;
A = [ cos( theta)  -sin( theta )
      sin( theta )  cos( theta ) ]
```

```
A = [ 2      1
      0      2 ]
```

```
A = [ 2      -1
      -1      0.5 ]
```

```
A = [ 2      1.5
      1      -0.5 ]
```

```
A = [ 2      -1
      1      0.5 ]
```

Can you explain some of what you observe?

☒ done



Solution

For a complete solution, see the [notes](#).

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